Interactive Media: Concept Development Template

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How I am using this document

This document is my own brainstorming before I create any code. This will help me decide what I want to do for my final assignment. All the sections in this are based off the assignment page and allow me to complete all the research that is required.

1. Brief selection

Record every brief you considered, then justify the one you chose. Your concept must sit inside a brief from the book, or it receives zero marks.

Brief	Core idea of the brief	Fit with my interests and skills (1 to 5)	Shortlisted?
Parametric object	Design a family of physical objects that can be controlled by parameters, usually intended for fabrication by 3D printing, laser cutting or plotting.	4	Yes
Data self-portrait	Collect data about myself and turn it into a visual portrait. The data can cover habits, movement, messages, sleep or screen time	5	Yes
Virtual creature	Create an artificial creature, or a small ecosystem of the creatures that seem alive through its movement and behaviour.	2	No
Generative landscape	Write a program that generates a landscape like a terrain or a city or a microscopic world or outer space. The output should vary each time it runs or unfold endlessly	3	No

Chosen brief: **Data self-portrait**

Why this brief over the others: I chose the data self-portrait because it lets me work with my own Spotify and Apple Music listening history and turn it into a portrait. Unlike the other briefs, the content is personal rather than just invented, which means the project also becomes a way to see my own data from a new angle. It gives me room to test what I can build whilst also learning about how my listening history and habits have changed over the years

2. Concept statement and key concerns

One-sentence concept: A data self-portrait where years of my Spotify and Apple Music listening history grow into a set of trees like rungs that the viewer can scrub through and hear, exploring how the music I listen and return to has changed over my lifetime.
Expanded description: Each artist will act as a tree in a garden. I am walking through the garden from left to right and as I go through time trees get planted. They grow flowers and wilt depending on if I listen to that artist in that period of time. As I walk through the garden the trees stay where they are and keep the same artist identity, which means I can let them flourish and die whenever I start and stop listening to that specific artist through time. I want to do it this way because not only can you see who you have been listening to at that time, but you can also see what date you started listening to them and how much you listen to them (based on how many flowers are on the tree).
I am thinking I will create this in weeks because days is too much and months is not enough detail for binging periods or periods of not listening to music.

Key concern	What it means in this project	How the work expresses it
1. Growing up with music	Taste changes over a decade but things that I once love stay part of me	Growth is permanent while the bloom is temporary, which mean spare artists are still visible even after they wilt. The timeline shows the taste vary from Ed Sheeran all the way to Geese.
2. No single company sees my entire history	Each platform holds only part of my listening, so each one's portrait of me is incomplete	Spotify and Apple data are merged into one garden, and the stacked timeline shows the switch between apps.

Test:

3. Rationale and influences

Rationale: I want to explore how a decade of possibly collected listening data can work as a self-portrait rather than as the year end ranking such as Spotify wrapped. Music is tied to specific periods of my life, so a long listening history that I can scroll through should show how my taste and identity have changed over time. Interactive media suits this because the data is already about sound, and a viewer can move through time instead of reading a static summary.

Influence (artist, work, theorist, technique)	Source / link	What I take from it	How it could appear in my concept
Dear Data, Giorgia Lupi and Stefanie Posavec (2014 to 2015)	Here	Over a year, each designer gathered one kind of data about their own life every week and hand drew it on a postcard to post to the other person. There were topics such as transport, physical contact and saying thank you. Small, everyday habits can say a lot about a person over a period of time.	Weekly data as the base unit and an organic handmade feel instead of taking standard chats. My work will also take the opposite approach to data collection. Dear Data is small and hand gathered, while my data is quite large and passive, as it has been recorded over a number of years.

Influence (artist, work, theorist, technique)	Source / link	What I take from it	How it could appear in my concept
Data Visualisation Self Portrait, Thabata Romanowski	Here	She based the piece on an exercise from Lupi and Posavec's project, giving each other personal trait is its own symbol and drew on modernist abstract artists such as Kandinsky and Klee. She later added annotations and a legend so viewers could decode the painting.	One consistent symbol system, where each artist becomes a plant with its own species, colour and shape. An abstract portrait still needs a way to decode it, which will be hover labels and artist information panels.
Themail, Fernanda Viegas (2006)	Here	Personal digital traces in this case years of email, can show how relationships change over time. Also shows how her as a person interacted with her email and how she used it.	Each artist treated as a relationship that grows, peaks and fades across the decade.
Favicon Diary, Shan Huang	Here	Passive digital records become a diary without any deliberate logging	My listening history will be used as an automatic diary of my life, without me having to log anything.

4. Precedent research and gap analysis

Existing work (creator, year)	Link	Visual approach	Audio approach	Interaction approach	Strength	Limitation or gap
Dear Data, Giorgia Lupi and Stefanie Posavec (2014 to 2015)	Here	Hand drawn post cards with a new visual system each week	None	None	Shows that smaller, everyday data collection can be very personal and expressive	Analogue, static and short term with the need to collect data.
Themail, Fernanda Viegas (2006)	Here	Words from email exchanges with each contact, laid out over time.	None	Select a contact and browse your shared history	Treats a digital record as a long-term relationship that changes over years	Its silent and focuses on one data type. Personal meaning still has to come from the user.
Favicon Diary, Shan Huang	Here	Site icons from browsing history	None	None	Turns passive digital traces into a	Silent and covers a single

Existing work (creator, year)	Link	Visual approach	Audio approach	Interaction approach	Strength	Limitation or gap
		arranged as a visual diary			diary without manual logging	platforms view of the person.
Spotify Wrapped	Spotify	Bold, branded story slides with top five rankings	Plays snippets of your top songs	Tap through slides and share to social media	Massive reach, to anyone that uses Spotify and uses real music as audio	One year only, and ranks rather than explores, resets every December and only sees Spotify listening.
Quotidian Record (Brian House, 2012)	Levin and Brain (2021), p.65	A vinyl record	A year of personal location data turned into music	None	Shows that personal data can be experiences as sound	Fixed and linear, so the listener can't explore, filter or move through time

Examples from the brief reviewed (Levin & Brain, 2021, pp.61 to 65):

- Shan Huang, Favicon Diary (2014): Passive browser data can become a portrait without any deliberate tracking
- Lupi and Posavec, Dear Data: small, specific data about daily life can be intimate and expressive
- Tracey Emin, Everyone I have ever slept with 1963-1995 (1995): a self-portrait can be a list of relationships that are made physical, as names embroided inside a tent
- Fernanda Viegas, Themail (2006): a digital archive can be read as a set of relationships rather than a list of messages
- Sonya Rapoport, Biorhythm Audience Participation Performance (1981): she compared her experiences with a machine's predictions about her, which questions whither computed data matches her real self.
- Hasan Elahi, Stay (2011): self-surveillance can be turned back on the watcher, since Elahi sends himself photographs of his daily life to the FBI himself

The gap or opportunity: Data self-portraits have used sound before, as in *Quotidian Record,* and digital traces before, as in *Themail* and *Favicon Diary*. None combine data from several platforms with free exploration for the user through time and sound, generated live from the data they are viewing. Evergreen will merge ten years of Spotify and Apple Music history into one interactive garden, so the viewer can move through their listening, hear it change and see the parts of themselves no single platform can record.

5. Media components

Component	What it is	How it expresses the key concern	Technique / library
Visual	A garden where each of my top 60 artists is a tree. Total hours set a plants size and number of branches, the recent listening stats bloom its flowers and abandoned artists droop, they become brown and drop petals. Seasons follow the southern hemisphere calendar, and the sun grows with how much I listened that week.	Concern 1: Growth never shrinks, but bloom comes and goes which means past artists stay visible even after they wilt and I'm not listening. Concern 2: A stacked timeline will show Spotify (green) and Apple music (pinkish red), making the switch between apps visible.	P5.js branching, seeded randomness (mulberry32) so each artist always gets the same species and colour, Perlin noise () for wind to let the petals drop naturally, a particle system for petals and a cached createGraphics timeline.
Audio	An ambient soundscape of six sine voices assigned to the most active plants, tuned to a pentatonic scale with trees low and wildflowers high. Obsession days trigger a nice chime, and clicking a plant plays that artist's top song	Concern 1: voices fade out as plants wilt, so abandoned artists become literally quieter. Concern 2: Real songs only play through Spotify. Apple music tracks have to open in Apple Music, so the audio itself is limited by which platform holds the data.	P5. sound (oscillator, envelope, reverb) and the Spotify Embed iFrame API, with the soundscape ducking to around 10% while a song plays.
Interaction	The viewer will scrub or play through 2016 to 2026, hover plants to hear a note, click plants to open their history and stats and jump to obsession days on the timeline.	Concern 1: moving through time lets the viewer watch their taste change. Concern 2: the proposed "Who's watching?" lens will switch between Spotify only, Apple only and combined gardens	Mouse hit testing on branch segments, a draggable timeline at the bottom, keyboard controls and a HTML panel on the side.

Integration: The timeline play head pretty much drives everything. As the user drags it, each artist's recent listening value changes, which sets how open its flowers are and how loud its voice is. A plant in full bloom is visible and audible at once, and a plant I've abandoned droops and goes quiet together. Obsession days connect all three components. The play head crossing makes the plant grow, bursts its petals and plays a chime.

Input	Visual response	Audio response
Drag or play the timeline by pressing the space bar	Plants grow, bloom and wilt, and seasons change and the sun size changes.	Voices are reassigned to the most active plants, so the soundscape shifts with my taste
Hovering over a plant	Artist's name appears and the plant has a highlighted circle around it	A soft pluck on that plants note
Click a plant	Side panel opens with first listen, hours, skip rate, % on Apple music and an obsession curve	The artists top songs plays and the soundscape ducks

6. User experience walkthrough

The user will experience the garden as a ten-year journey of my Spotify and Apple music history. At the default speed of five seconds per week, the full decade plays in under two minutes, so a complete viewing fits inside a short time.

Stage	What the user sees	What the user hears	What the user does	What they should feel or understand
First encounter	An intro screen with some brief information and introduction	Silence, since browsers block audio until the first click	Clicks to enter, which also unlocks sound	Curiosity, and a clear question to hold while watching
Discovery	July 2016: an almost empty garden with a few small plants that looks like they've just been planted. They start to grow and the timeline starts to play.	The first sine voices fade in as the early plants come alive	Watches, then starts hovering over plants which shows the name and plays soft notes.	The garden is made of artists, and it changes as time passes
Exploration	Plants growing, blooming and wilting as the years pass, with seasons shifting and the sun swelling in heavy listening weeks	The soundscape moving between voices as different artists take over	Click plants to open their history, drags the timeline back and forth and pauses on interesting moments	Taste isn't fixed, and each plant has its own story of discovery of a specific artists and obsession
Peak or turning point	14 July 2023 the Billie Eilish plant glows, 90 petals burst and a caption explains the 159 plays of "What Was I Made For?" in one day. Later, in May 2026 the timeline turns from green to pink as Spotify listening drops, and I change to Apple Music	A rising chime on the obsession day, then the soundscape continuing as Apple Music takes over. No sound for the difference in streaming services.	Clicks the obsession diamond to replay it and notices the platform switch on the timeline	Data hides real moments in a life, and one apps record would make me seem to stop listening altogether
End state	September 2026: a fully grown garden with current artists such as Olivia Dean and Geese bloom, while early favourites like Ed Sheeran and Vance Joy are still there but have no flowers. Billie Eilish is the largest tree	A soundscape made of the music I'm listening to now	Explores freely, comparing plants or scrubbing the timeline back to past dates	My taste grew has grown up, but what I outgrew in terms of music still shaped me, signified by the plants still being visible.

How does the user learn to interact without instructions?

- Auto-play shows change happening straight away, so the viewer understands that time matters before touching anything
- Hover feedback gives every plant a name and sound, which lets the user click
- Visible landmarks on the timeline, the pink obsession diamonds show where the important moments are
- A one-line key on the intro screen explaining the core mapping. For example, “Plants grow with every hour I’ve listening. They bloom while I’m listening and die when I stop”. Without it the viewers will see change but not necessarily know what bloom and wilt signify.

7. Technical research: libraries and techniques

List the tools you investigated and justify each one by what it does for the concept, not how impressive it looks.

Library or technique	What it does	Why it serves my concept	Tutorial / docs link	Tested?	Use it?
P5.js.1.11	Core drawing, animation and output	Required by the subject, and its HSB colour mode makes season, and wilt colour shifts simple	P5js.org/reference	Yes	yes
P5. sound	Oscillators, envelopes and reverb	Enough to build a soft, data driven soundscape without adding an entire other library	P5.js reference, p5. sound section	Yes	Yes
Tone.js	Advanced synthesis, scheduling and effects	Could give richer sound, but p5 sound already covers six voices, plucks and chimes which is all I need	tonejs.github.io	Yes	No, it adds complexity for no reason
ml5.js (handPose, bodyPose, faceMesh)	Machine learning, body hand and face tracking	No body input connects to listening history, so gesture control would be a gimmick. Could put in later if there’s time to be able to control timeline with hands	Ml5js.org	No	Maybe
WEBGL mode / shaders	3D rendering and GPU effects	The garden is 2D and illustrative, and face perspective achieves depth more cheaply	P5.js reference	No	No

Library or technique	What it does	Why it serves my concept	Tutorial / docs link	Tested?	Use it?
matter.js (physics)	Physics engine	Falling petals only need gravity and drift, which a custom particle system handle	Brm.io/matter-js	Researched	No
Recursive branching	Builds tree like plants by repeating a branch rule	Plant size and recursion depth (1-6) map directly to total hours of an artist	The Nature of Code, Daniel Shiffman	Yes	Yes
Seeded randomness (mulberry32)	Random numbers that repeat for the same seed	Each artist always gets the same species, colour and position so the garden stays specific for each viewing	Here	Yes	Yes
Perlin noise / flow fields / particles	Smooth, natural randomness	Wins sway and petal drift make the garden look more real than mechanical	P5.js reference	Yes	Yes
Particle system	Many small moving objects with simple physics	Falling petals show wilting and obsession burst	The Nature of Code	Yes	Yes
Spotify Embed iFrame API	Embeds a playable Spotify track	Let's the viewer actually hear the real song that I have most listened to from that artist	Here	Yes	Yes
Python data pipeline	Cleans and merges and aggregates the exports at once	Merging two platforms into one record is important for concern 2, and doing it offline keeps the sketch fast	Scott Murray, Creating Coding and Data Visualisation with p5.js (2017)	Yes	Yes
createGraphics and drawingContext	Off-screen buffers and native canvas gradients	Caches the timeline chart and draws soft gradients while holding 60fps	P5.js reference	Yes	Yes
Manrope (google font)	Typeface	A clean and friendly font used in the garden	Fonts.google.com	Yes	yes

Technical risks identified:

- The sketch needs a local server. Opening the index.html wont load garden. Json
- Browser audio is blocked until eh user interacts with the page
- Spotify playback is limited. The browser can only play 30 second previews and needs an internet connection
- Apple data is daily, with no time of day so the two sources don't match in detail
- Artists names may not match which could split one artists into two plants

Mitigation for each risk:

- Server: ship the data as a .js file so the sketch opens by double clicking or keep Live Server ready for the presentation
- Audio: the enter the garden clicks double and the audio unlocks
- Spotify: the soundscape works without Spotify
- Name matching: match the exact name first, then the first credited artist and check top 60 by hand
- Extreme values: scale growth with a power curve and cap the bloom so the obsessions cant dominate the garden and look unrealistic
- Privacy: strip IP and device data in the pipeline, and leave the raw exports out of the submitted code folder
- Performance: precompute all data series at load, cap particles at 500 and cache the timeline

8. Scope, originality and feasibility

In scope (core features for the draft)	Links to key concern	Out of scope (stretch or cut)	Reason for cutting
Data pipeline merging Spotify and apple music into one weekly dataset	Concern 2	Travel postcards (cut)	Country data relates to neither concert and includes VPN noise when used
Garden of 60 plants that grow bloom and wilt	Concern 1	Search (cut)	A convenience, not part of the portrait
Timeline with play, scrub and a stacked Spotify/apple music chart	Concerns 1 and 2	Poster export (cut)	A convenience and a static output works against an interactive concept
Data driven soundscape tied to the most active plants	Concern 1	Fireflies for late night listening (cut)	Apple data has no time of day so this would present Spotify partial view as complete, which contradicts concern 2
Hover notes and click to open artist panel	Concern 1	Skip rate wind (cut)	Adds visual noise without a clear link to either concern

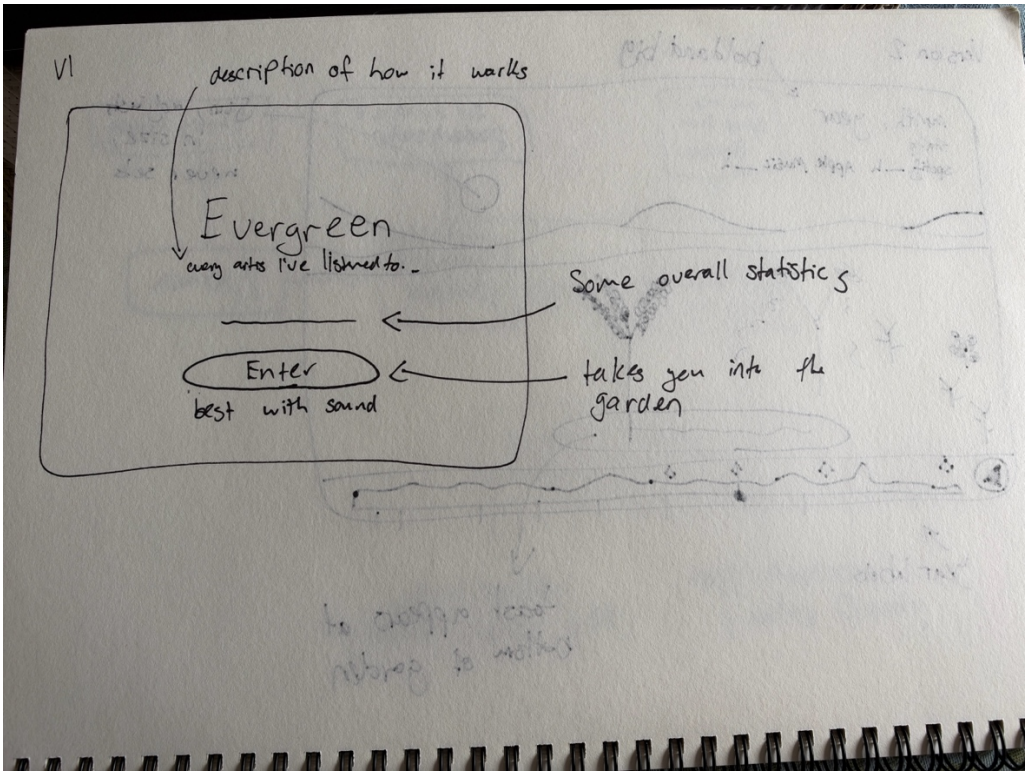
In scope (core features for the draft)	Links to key concern	Out of scope (stretch or cut)	Reason for cutting
Obsession days with glow, petals, chime and caption	Concern 1	Soundtrack through time (stretch)	Serves concern 1, but depends on Spotify premium and would compete with the soundscape
Top song playback through the Spotify embed	Concerns 1 and 2	“Who’s watching?” lens (maybe stretch for A2B)	Not needed for the draft but could be used in the final submission

Original components:

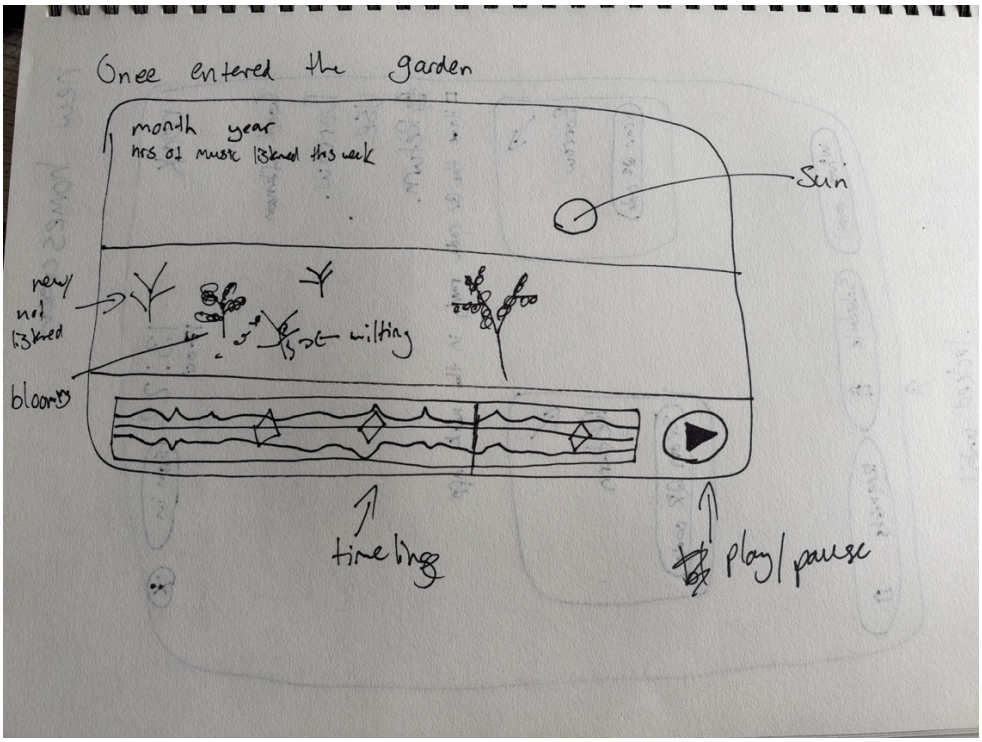
- Cross-platform merge: a pipeline that reconciles two export formats, resolves artist’s names between the two service s and strips any private data
- Growth and vitality model: growth is permanent while bloom decays with a six-week half life, and only plants that were once loved can wilt
- Plant generation: each artist gets an identity through a tree
- Soundscape allocation: six voices are reassigned several times a second to the most active plants, which is tuned to a pentatonic scale by plant type
- Obsession detection and notes: the pipeline flags any day with 10 or more plays of one song and the sketch triggers when the play head crosses that week

Key point of distinction: unlike the examples that I researched, Evergreen will merge data from two platforms into one portrait that the viewer can explore through time while hearing it change, which shows that no single company’s data captures something so large.

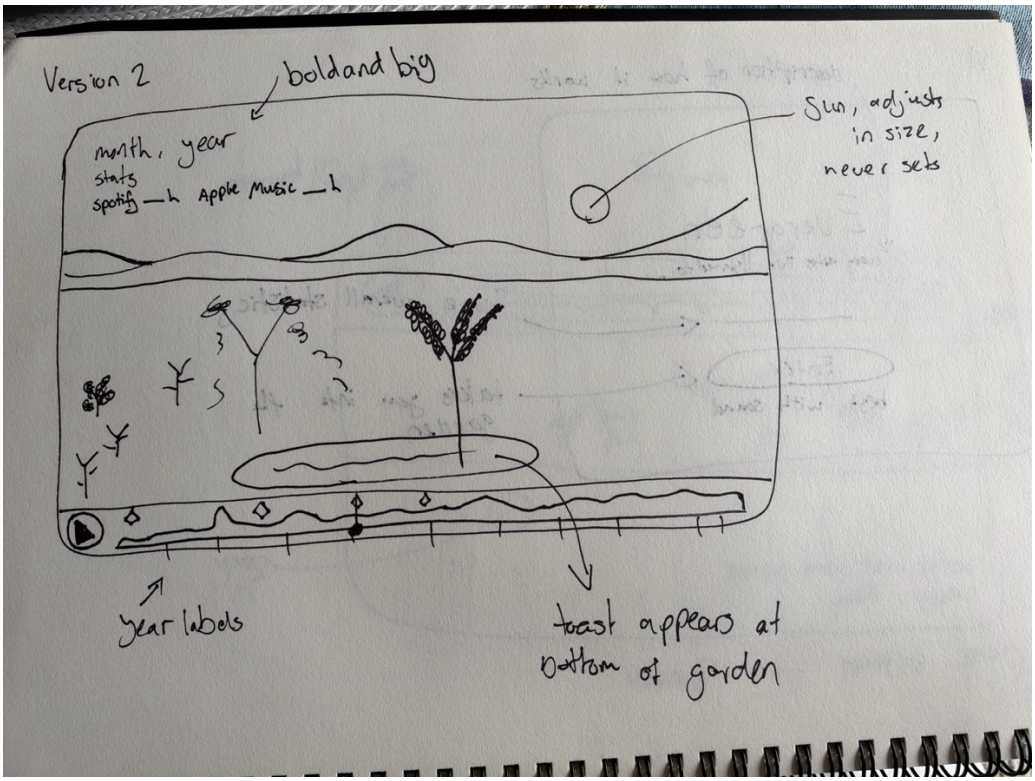
9. Sketches and system diagrams



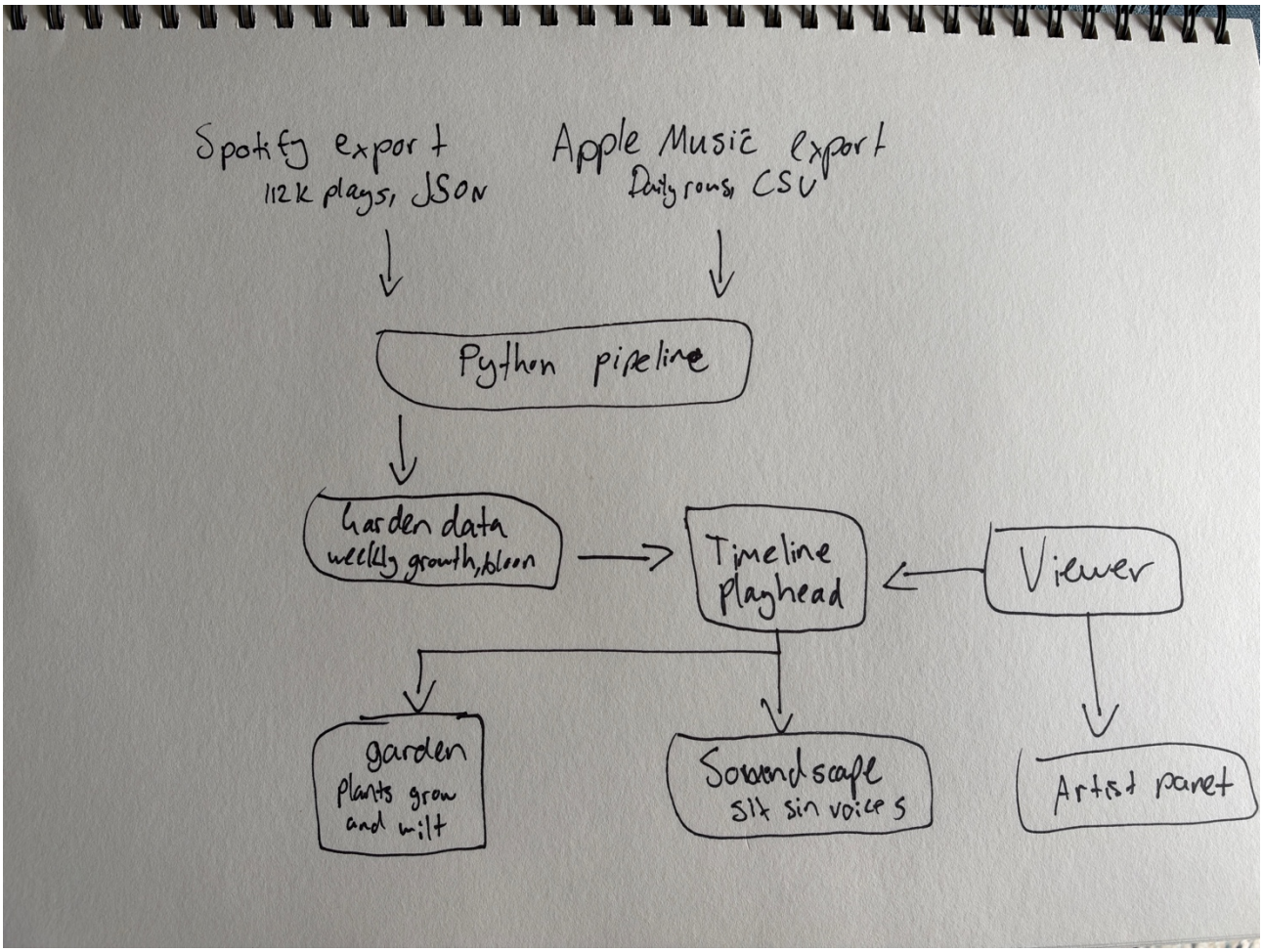
Introduction screen sketch



Version 1 main screen very sketch



Version 2 less rough sketch with some changes



Processes table

10. Tutor feedback from draft submission

Date	From (tutor / peer)	Feedback	My response or change

Slide	Content	Draws from sections	Rubric criteria covered
1	Concept, rationale, influences, gap	1, 2, 3, 4	
2	System diagram, media components, interaction, originality	5, 6, 7, 8, 9	
3	Draft status, feedback, adjustments, next steps	10 + draft notes	

- ☐ Every rubric criterion appears in the right-hand column at least once
- ☐ Sketches and diagrams are exported as images for the slides